AMERICAN UNIVERSITY OF ARMENIA
ENGINEERING SCIENCES

Feasibility of Scientific Balloon Navigation using Angle and Speed Control

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Aim of study

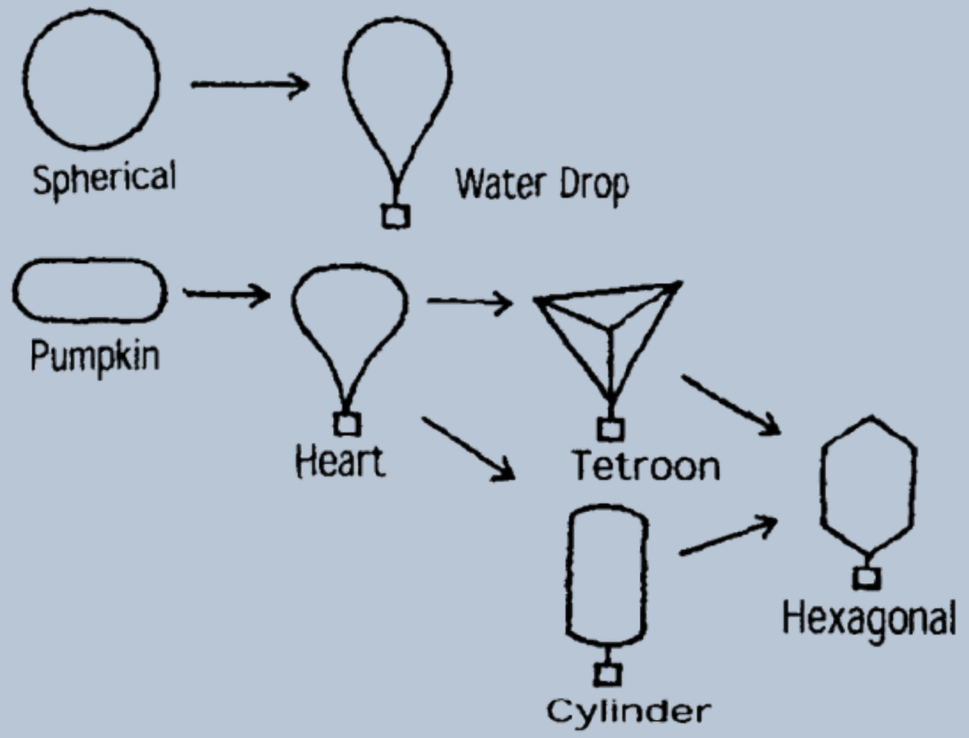
The primary aim of this study is to design and analyze a controllable high-altitude scientific balloon system capable of maintaining orientation and stability during flight. The system integrates:

- Angular control using a reaction wheel and decoupler.
- Directional control using a brushless DC motor to counteract wind drift.

Through wind data analysis, control algorithm design, and system simulation, the goal is to assess flight feasibility and determine optimal operating conditions—particularly identifying the altitude and timing where wind speeds are minimal for stable operation.



Literature Review Summary



Types of high-altitude balloons:

Rubber, stratospheric

Shapes and pressure types

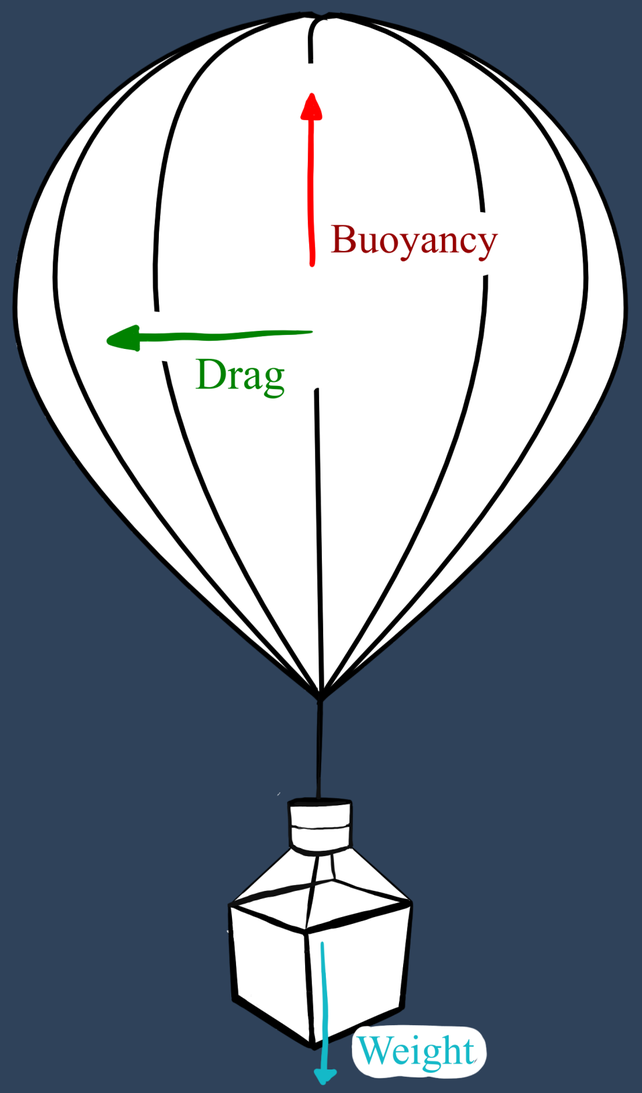
Spherical Balloons, Natural Shape Balloons, Heart-shaped balloon, Water-Drop Shaped Balloons

Literature Review Summary

$$W_B = m_t \cdot g$$

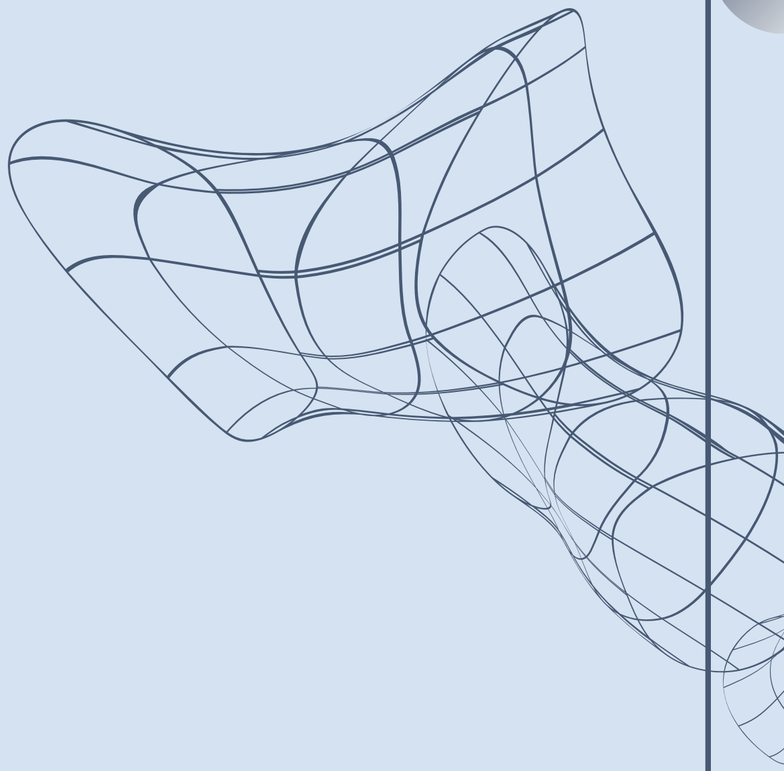
$$F_B = \rho_a \cdot g \cdot V_b$$

$$F_D = \frac{1}{2}\rho_a |\boldsymbol{v_{rel}}|^2 C_D A_b$$



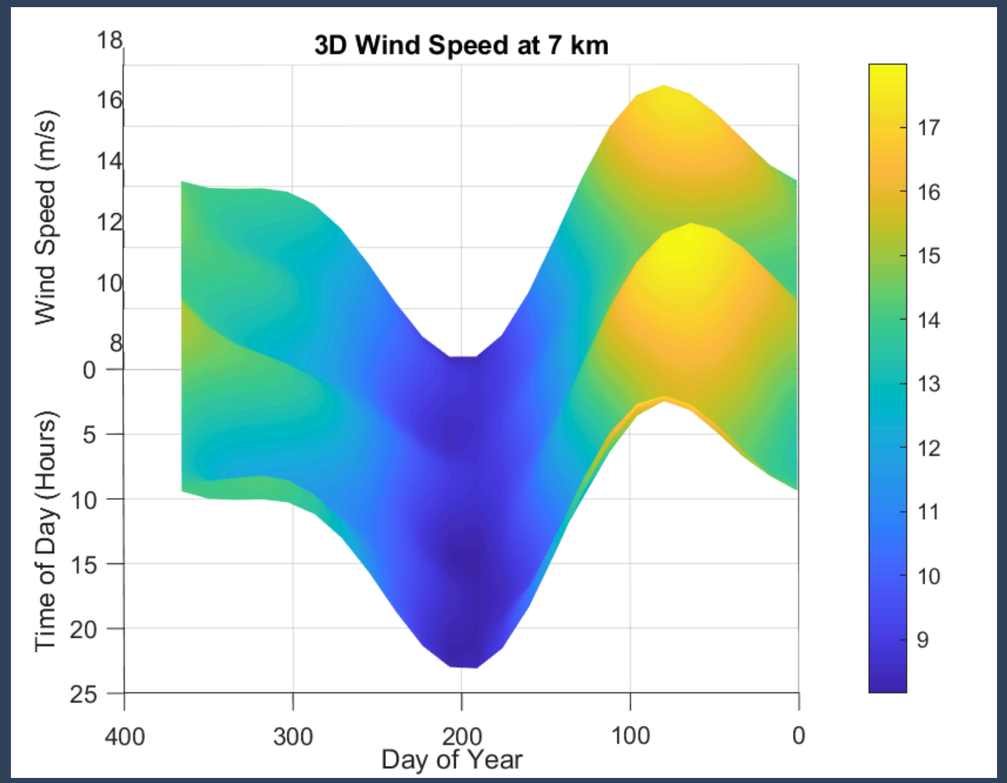
Wind Analysis

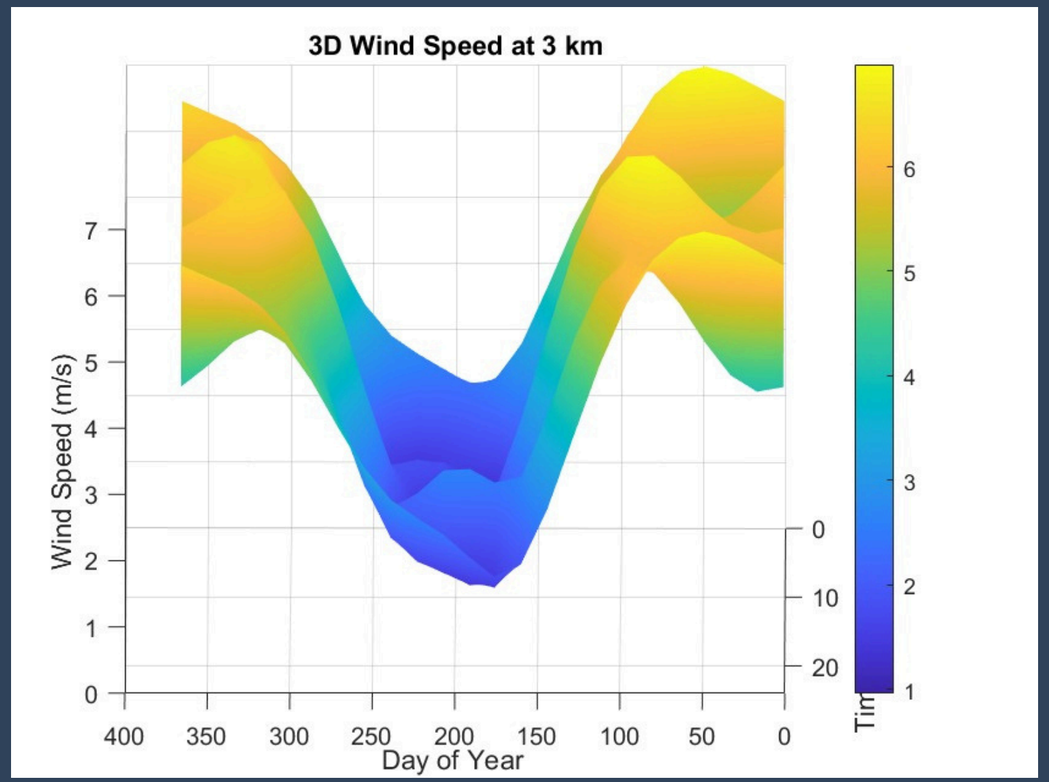
- MATLAB simulation across altitudes and coordinates
- Identify best flight window and altitudes (e.g., Day 210)
- Show wind variation plots
- Importance: system needs low-wind windows



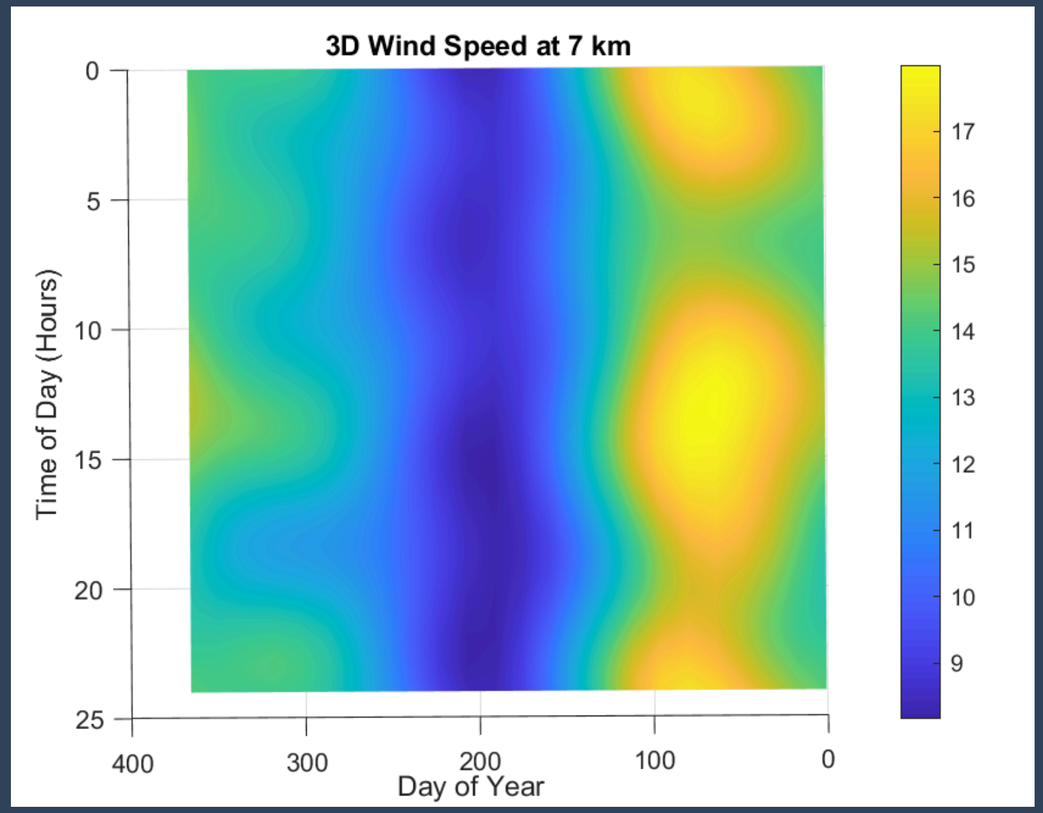
```
wind = atmoshwm(latitude,longitude,altitude,Name,Value)
```

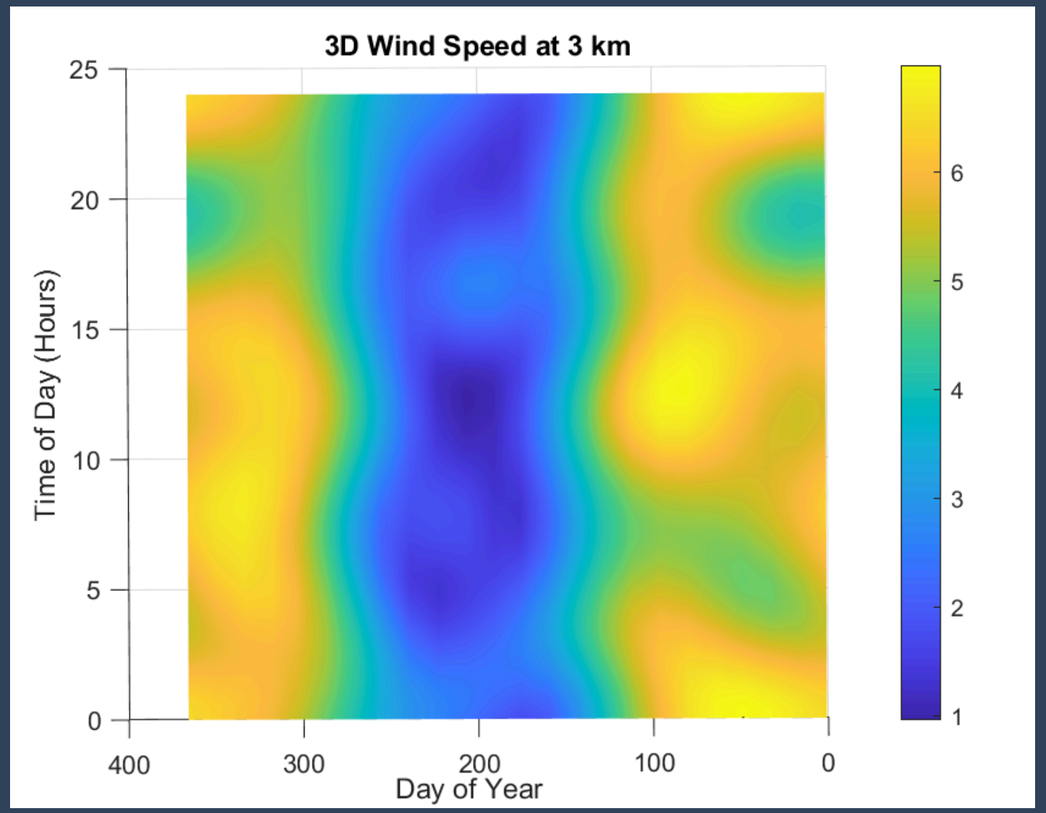
Atmospheric wind analysis using MATLAB (horizontal wind model)



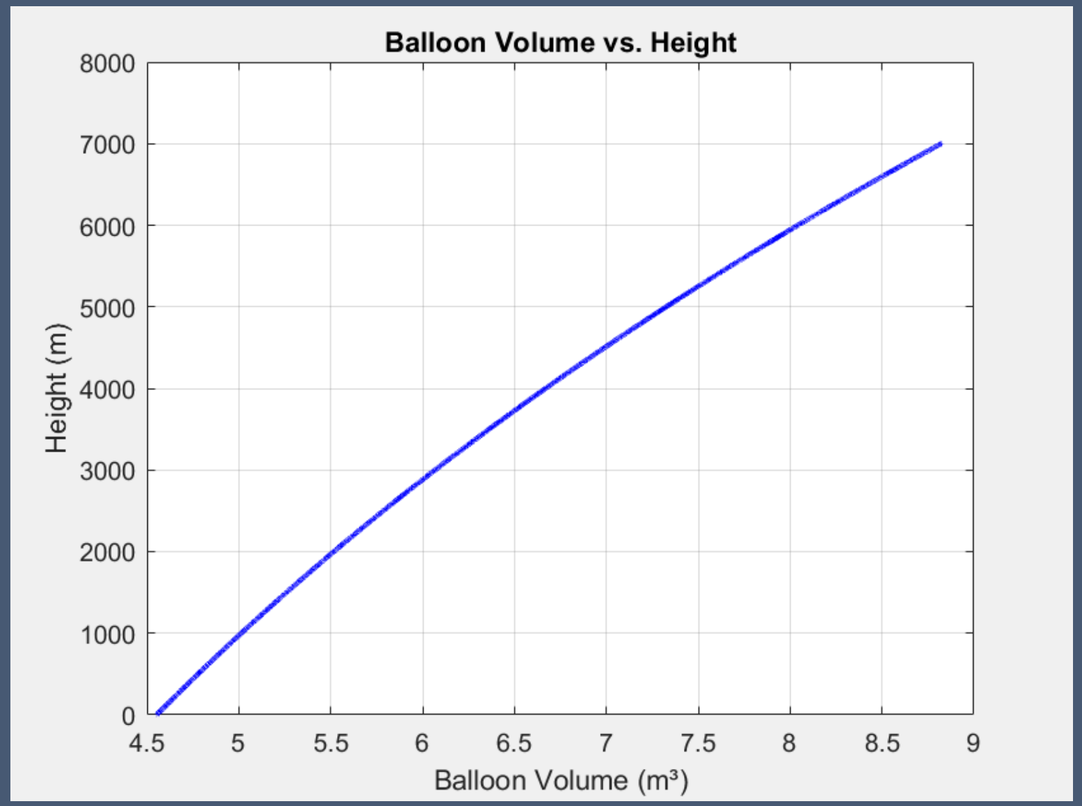


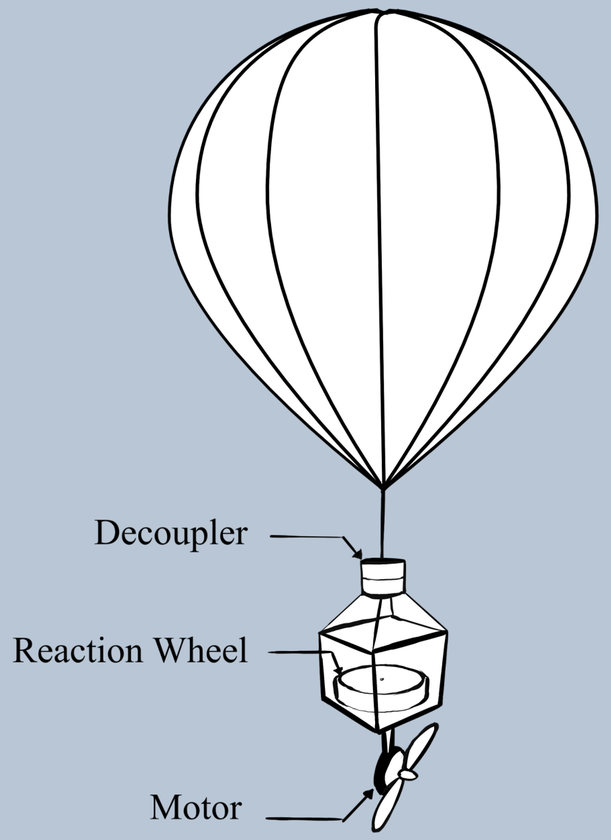
Atmospheric wind analysis using MATLAB (horizontal wind model)





Change of volume with Altitude





System Design Overview

Architecture:

Motor + Decoupler + Reaction Wheel

Directional control with a brushless DC motor fixed beneath the gondola. Angular control using a reaction wheel for steering the gondola’s angle and a decoupler, which separates the gondola from the balloon.

Three Parameter Optimization

The Volume of the balloon is dependent on the gondola mass, which is dependent on the thrust of the motor, which is dependent on the volume, etc...

Volume in terms of gondola mass

$$V_{b,0} = \frac{T(0)P(h)}{T(h)P(0)\rho_a - T(h)P(0)\rho_{He,surface}} \cdot m_g$$

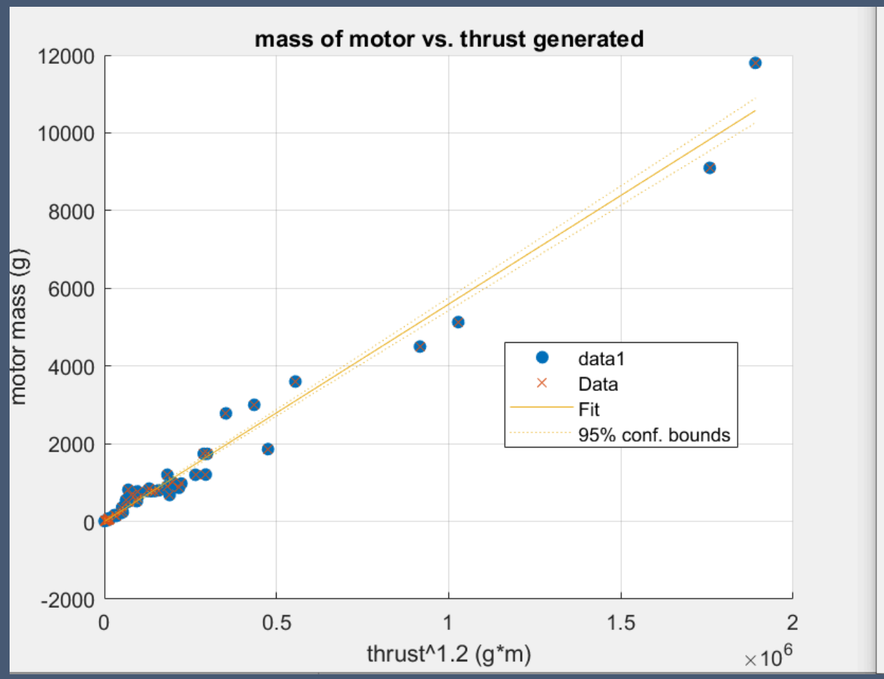
Motor mass in terms of thrust

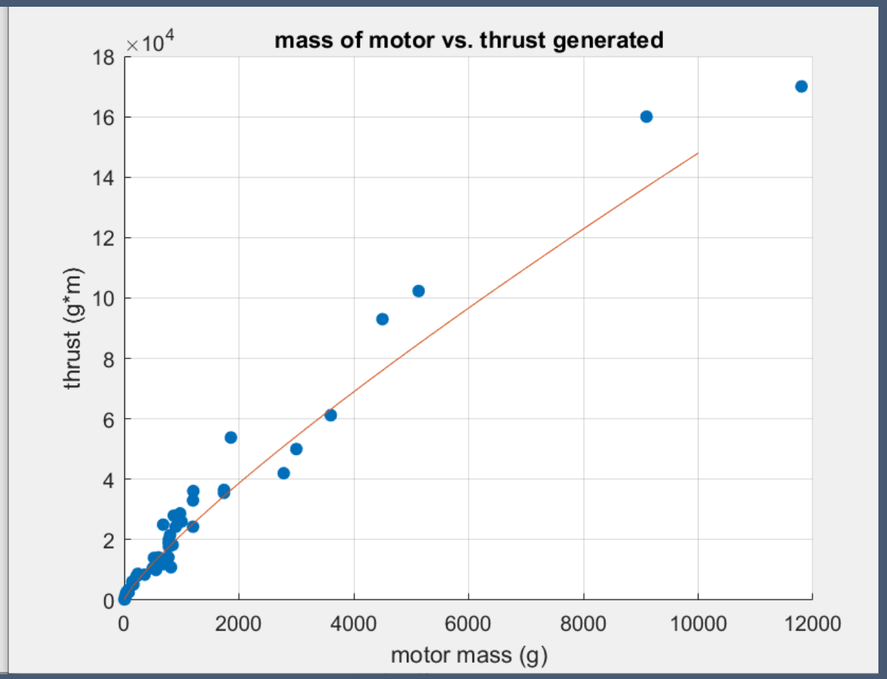
$$m_{\text{motor}} = a \cdot \left(F_T^{1.2}\right)$$

Thrust in terms of volume

$$F_{T,0} = c\frac{1}{2}\rho_0 C_D u^2 \pi \left(\frac{3}{4\pi}\frac{T(h)P(0)}{T(0)P(h)}\right)^{2/3} V_{b,0}^{2/3}$$

Mass to Thrust relationship for brushless motors





Three Parameter Optimization Example

Iteration	Gondola Mass (kg)	Initial Volume (m^3)	Rated Thrust of the Motor (N)
1	3.50	3.39	145.63
2	4.03	3.90	159.94
3	4.33	4.19	167.80
4	4.50	4.36	172.11
5	4.59	4.45	174.47
6	4.64	4.49	175.75
7	4.67	4.52	176.45
8	4.68	4.54	176.83
9	4.69	4.54	177.04
10	4.70	4.55	177.15
11	4.70	4.55	177.21
12	4.70	4.55	177.25
13	4.70	4.55	177.27
14	4.70	4.55	177.28

Control Strategy – PID Controllers

Two controllers were used:

one for controlling the motor thrust based on wind disturbances, and one for adjusting the gondola’s angle using the reaction wheel.

The controllers work independently

but are tuned to provide a stable and responsive control loop.

Simulation Results

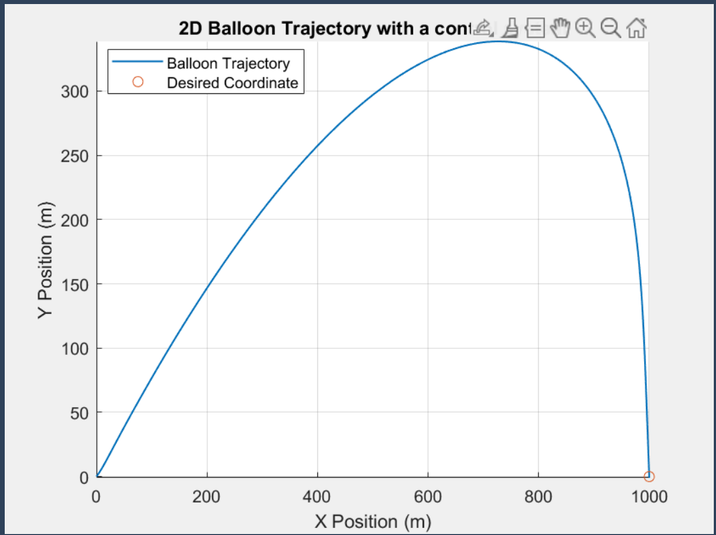
Altitude of 7000m

Initial Conditions:

r = 0, **v** = 0, θ = 0

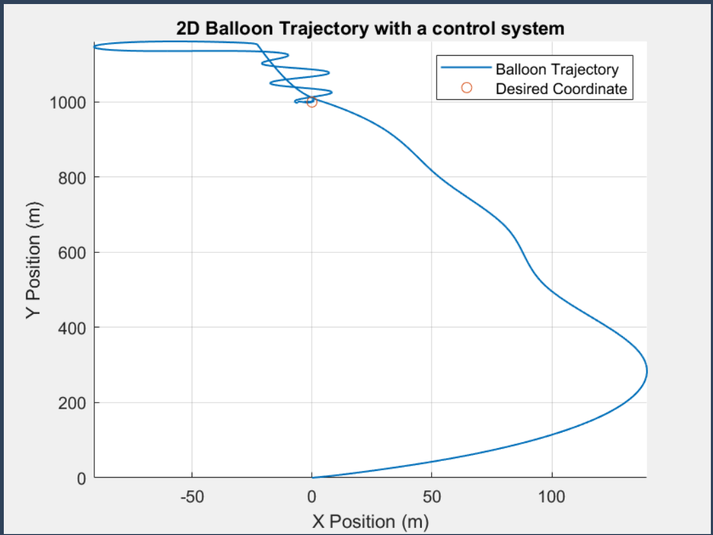
Kpm = 0.5, Kim = 12, Kdm = 6

Kpa = 0.07, Kda = 0.04



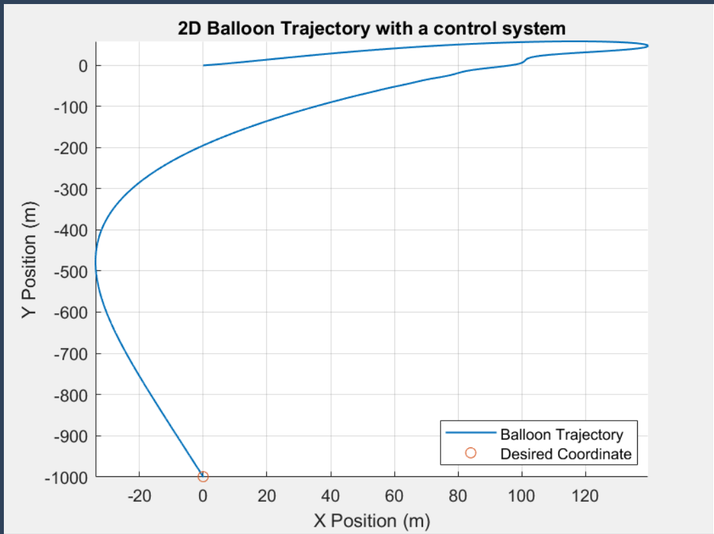
t = 246.5s, or about 4.1 mins

Vavg = 4.06 m/s



t = 168.1s or about 2.8 mins

Vavg = 5.95 m/s



t = 450.5s or 7.5 mins

Vavg = 2.22 m/s

Simulation Results

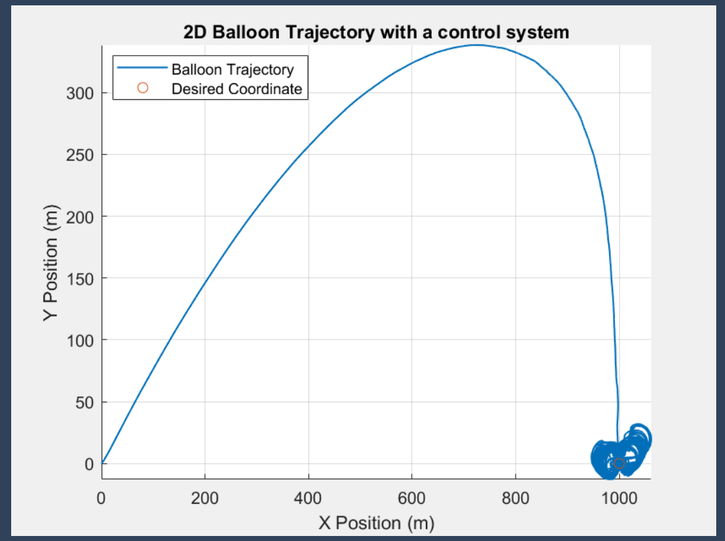
Altitude of 7000m

Initial Conditions:

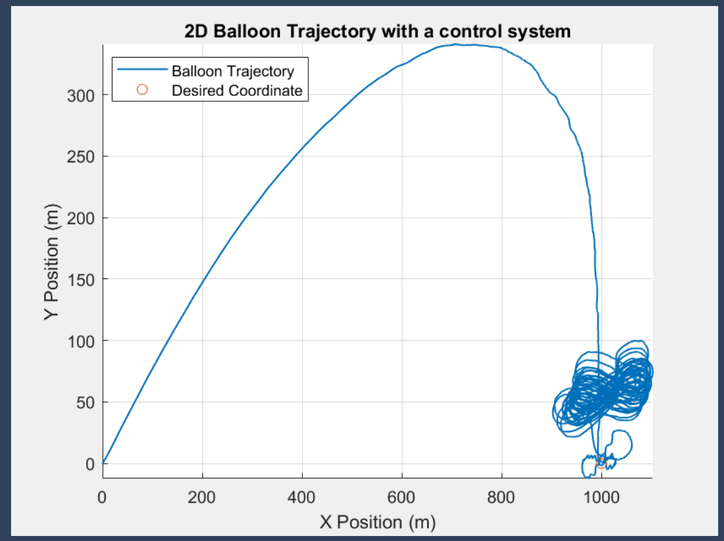
r = 0, **v** = 0, θ = 0

Kpm = 3, Kim = 15, Kdm = 6

Kpw = 0.6, Kdw = 0.3



Wind disturbance of +- 1 m/s



Wind disturbance of +- 2.5 m/s

Conclusion

- Successful modeling of balloon physics & controls
- Developed a working simulation and control al gorithm
- Main limitation: weight-to-payload ratio

Future Work

- fully automated system
- incorporate solar power subsystems
- descent of the balloon